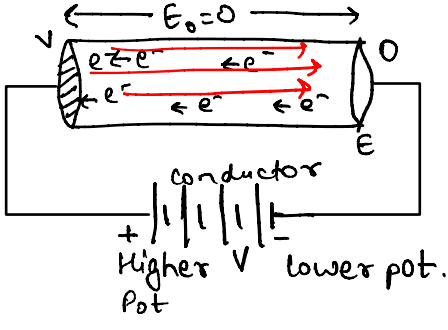
12th BOARD HACKER

Current Electricity

Electric current – Rate of flow of charge through wire

- conductors ⊖ e⁻ e⁻ e⁻ ⊖ free e⁻
  ⊕ charges are fixed in conductor
- current will be due to e⁻ only
- when there is no battery all e⁻ are in random motion
- no net current in conductor
- when battery is apply, there is force on e⁻ & they start moving towards (+) terminal.



$$I = \frac{\text{charge flow}}{\text{time}} = \frac{Q}{t} \quad \text{unit} = A$$

dimension = [A]

$$I = \frac{dq}{dt}, \quad \int dq = \int I \, dt$$

- electric field outside a current carrying conductor is zero but inside a conductor is V/L

- $I = (2+3t)A$. charge crossed in first 10 seconds

$$dq = I.dt$$

$$q_T = \int_0^{10} (2+3t).dt = 2[t]_0^{10} + 3\left[\frac{t^2}{2}\right]_0^{10}$$

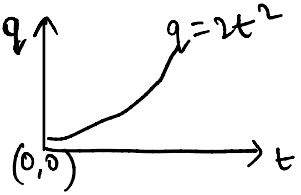
$$= 2(10-0) + \frac{3}{2}(10^2-0^2)$$

$$= 20 + \frac{3}{2} \times 100 = \boxed{170C}$$

- $q = 2t^2$

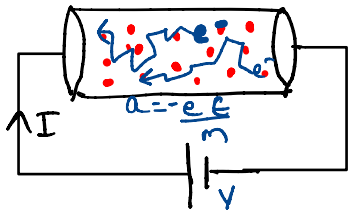
$$I = \frac{dq}{dt} = \frac{d(2t^2)}{dt} = 4t$$

$$I = 4t \quad \text{at } t = 3 \Rightarrow I = 12A$$



Drift velocity & Mobility — avg. velocity with which e^- drift towards the positive terminal of battery.

- Let there be battery of voltage V is applied across conductor of length 'L'.



- all e^- experience force

$$\vec{F} = -e\vec{E}$$

$$a = -e\vec{E}/m$$

- due to force all e^- experiences collision with atoms & drift towards ⊕ terminal of battery.

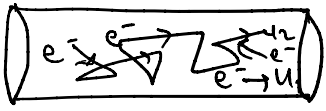
$$V_{drift} = \vec{V}_{avg} = \frac{\vec{v_1} + \vec{v_2} + \vec{v_3} + \cdots \vec{v_n}}{n}$$

$$= \frac{\vec{u_1} + \vec{u_2} + \cdots \vec{u_n}}{n} - \frac{e\vec{E}}{m}\left(\frac{t_1 + t_2 + t_3 + \cdots}{n}\right)$$

($\vec{u_1} + \vec{u_2} + \cdots \vec{u_n} \to 0$)

$$V_{drift} = \vec{V}_{avg} = \frac{-e\vec{E}}{m}\tau$$

τ = avg. time for collision = relaxation time



- e^- are in random motion
- there is no net flow of charge

Let e^- have initial velocity $\vec{u_1}, \vec{u_2}, \vec{u_3} \cdots \vec{u_n}$

$$\vec{u}_{avg} = \frac{\vec{u_1} + \vec{u_2} + \vec{u_3} + \cdots \vec{u_n}}{n} = 0$$

velocity of all e^- at time t.

$$\vec{v_1} = \vec{u_1} - \frac{e\vec{E}}{m}t_1$$

$$\vec{v_2} = \vec{u_2} - \frac{e\vec{E}}{m}t_2$$

$$\vec{v_n} = \vec{u_n} - \frac{e\vec{E}}{m}t_n.$$

Mobility– drift velocity attained per unit Ef.

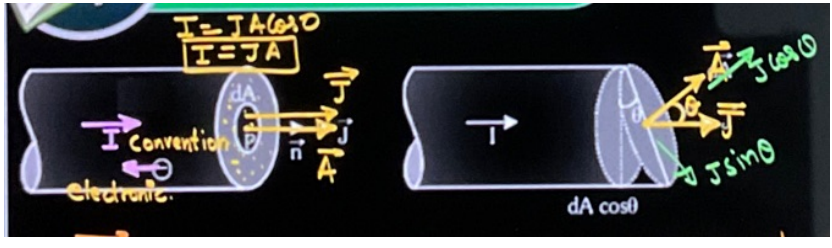
$$\mu = \frac{v_d}{E} = \frac{eE\tau}{mE} = \frac{e\tau}{m}$$

$m \downarrow \ \mu \uparrow$

$\mu_e > \mu_p$

$\mu_e > \mu_n$.

Current density



⊕ flow dirⁿ → I (conventional)
⊖ flow dirⁿ → opp. to dirⁿ of I.

$\vec{J}$ = current density = amount of current passing per unit area.

$\vec{J} = \frac{I}{A}$ unit = $\frac{A}{m^2}$ Dim = $[AL^{-2}]$

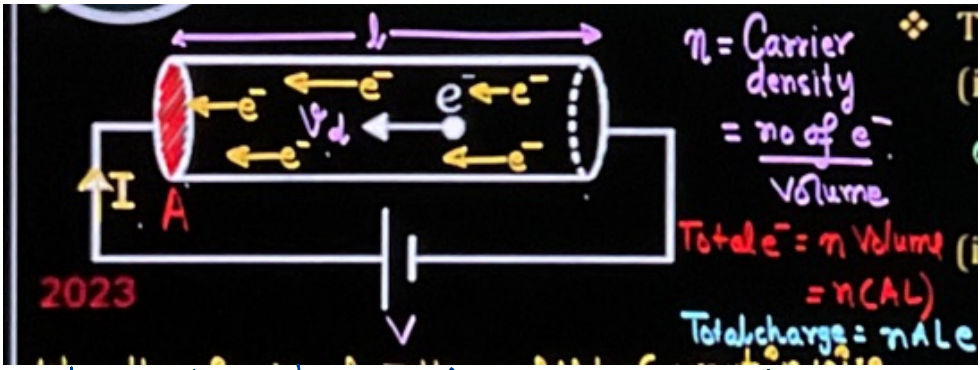
$\vec{J}$ is a vector quantity (dirⁿ is same as that of current)

∴ other form = $\vec{J} \cdot \vec{A} = I$ $\boxed{I = JA\cos\theta}$

example- $\vec{J} = (2 \times 10^4 \hat{j})\ Am^{-2}$
find the current through area $\vec{S} = (2\hat{i} + 3\hat{j})\ cm^2$

sol^n - $\vec{J} = (2\times10^4)\hat{j}\ Am^{-2}$
$\vec{A} = (2\hat{i} + 3\hat{j})\ (10^{-2})^2\ m^2$
$I = \vec{J} \cdot \vec{A}$
$(2\times10^{-4}) \times 0 + 3\times10^{-4} \times 2\times10^4$
$\boxed{I = 6A.}$

Relation between current and drift velocity



when there is V_d of e⁻ there will be current in wire

$I = \frac{\text{charge}}{\text{time}} = \frac{nALe}{t}$

$\boxed{I = neAV_d}$

V_d → drift velocity.

e^- charge

A → area of cross-section

n → carrier density $= \frac{\text{no. of } e^-}{\text{volume}}$ → depends on nature of cond.

$\text{speed} = \frac{d}{t}$

$V_d = \frac{L}{t}$

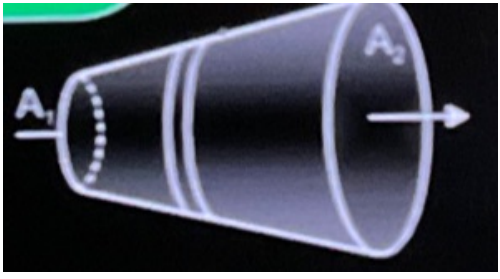
* The conductor remains uncharged.

i) It is same along the wire

$I_1 = I_2$

$I = neAV_d$

ii) Current density, electric field strength, drift velocity are inversely proportional to area. Here $I_1 = I_2$ but $A_1 < A_2$, so $J_1 > J_2$, $E_1 > E_2$, $V_{d_1} > V_{d_2}$.



conductor of random shape

$AV_d = \text{const.}$ $\quad J = \frac{I}{A}$ (same)

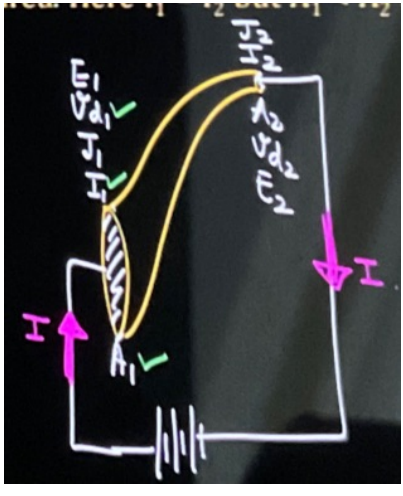
$\vec{J} = \sigma \vec{E}$

$A_1 > A_2$

$V_{d_1} < V_{d_2}$

$J_1 < J_2$

$E_1 < E_2$.



Ohm's Law, Resistance, Resistivity

↳ when we apply battery of voltage V across a conductor, relation between V & I.

$$E = \frac{V}{L}$$

$I = neAv_d$

$$I = neA\frac{eEt}{m}$$

$$I = \frac{ne^2\tau AE}{m}$$

$$I = I\left(\frac{ml}{ne^2\tau A}\right)$$

Resistance, $R = \frac{ml}{ne^2\tau A}$

length of conductor $\rightarrow R \propto l$

area of cross section $\rightarrow R \propto \frac{1}{A}$

nature of Mat. $\rightarrow R \propto \frac{1}{n}$

Temp. $\rightarrow R \propto \frac{1}{\tau}$

Ohm's law –

at low & const. temp. $V_{app} \propto I_{\text{through cond.}}$

resistivity – depends on temperature

– temp ↑ – collision rate ↑ – time b/w collision

– τ = avg. time of collision ↓ ⇒ R ↑.

$\boxed{R = R_0(1+\alpha\Delta T)}$

R_0 at 0°c; α ↳ temp. coeff. of resistance; ΔT ↳ change in temp.

$$\boxed{R = \rho\frac{l}{A}, \quad \rho = \rho_0(1+\alpha\Delta T)}$$

Microscopic Ohm's law vector form law

from relation – 1 $\frac{I}{A} = \frac{ne^2\tau E}{m}$, $\vec{J} = \frac{\vec{E}}{\rho} \Rightarrow \vec{J} = \sigma\vec{E}$

$\sigma = \frac{1}{\rho}$ = conductivity.

(property of cord by which it opposes flow of current through it) unit = Ω ohm

$R = \frac{V}{I} = [ml^2T^{-3}A^{-2}]$

$V \propto I \Rightarrow V = IR$

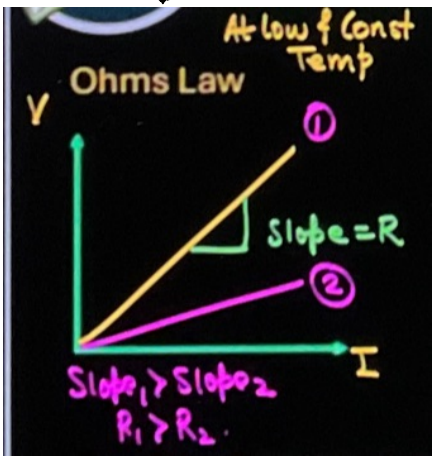
$R = \frac{V}{I}$

conductance $\Rightarrow G = \frac{1}{R}$

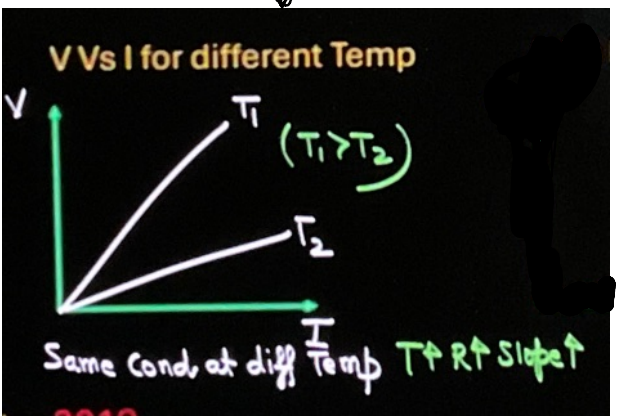
unit = Ω^{-1}, mho, siemen, s.

Important graphs

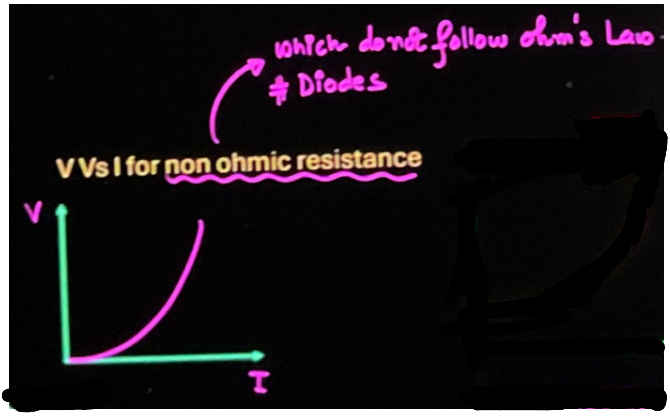
Ohm's Law



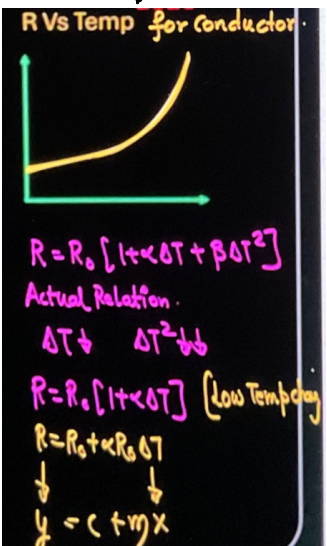
V vs I for diff. temp.



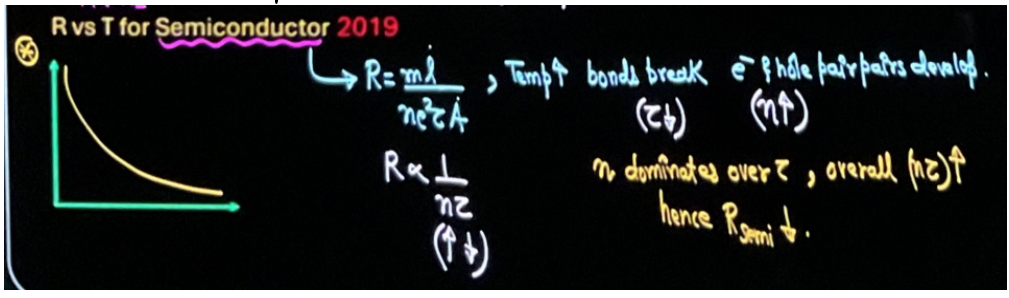
V vs I for non-ohmic resistance



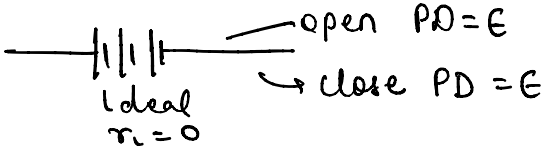
R Vs temp for conductor

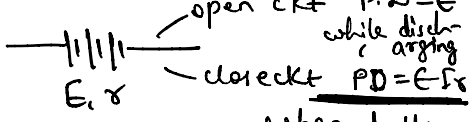


R Vs T for semiconductor



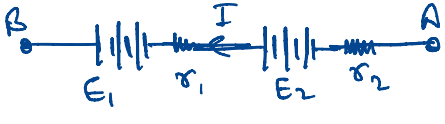
Combination of Cells

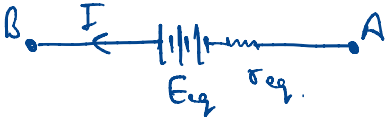




when battery is charge
$PD = \varepsilon + Ir$

a) Series combination





$V_A - Ir_2 + \varepsilon_2 - Ir_1 + \varepsilon_1 = V_B$

$$\varepsilon_2 + \varepsilon_1 - I(r_1 + r_2) = V_B - V_A \quad \rightarrow (1)$$

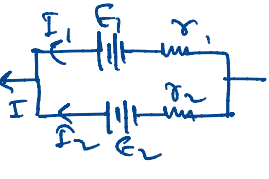
$V_A - Ir_{eq} + \varepsilon = V_B$

$$\varepsilon_{eq} - Ir_{eq} = V_B - V_A \quad -(2)$$

b) Parallel combination

$I = I_1 + I_2$

$$\frac{\varepsilon_{eq}}{r_{eq}} = \frac{\varepsilon_1}{r_1} + \frac{\varepsilon_2}{r_2}$$



If there are n batteries

$$\frac{\varepsilon_{eq}}{r_{eq}} = \frac{\varepsilon_1}{r_1} + \frac{\varepsilon_2}{r_2} + \frac{\varepsilon_3}{r_3} + \cdots$$



on comparing 1 & 2

$E_{eq} = E_1 + E_2 \quad r_{eq} = r_1 + r_2$

If n batteries tnt

$E_{eq} = E_1 + E_2 + E_3 + \cdots E_n$

$r_{eq} = r_1 + r_2 + r_3 + \cdots r_n$

In n equal battery

$E_{eq} = nE \quad r_{eq} = nr$

n identical batteries in parallel.

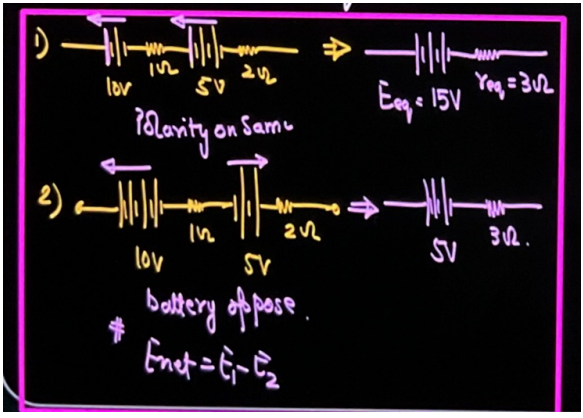
$\frac{1}{r_{eq}} = \frac{1}{r} + \frac{1}{r} + \frac{1}{r} + \cdots$

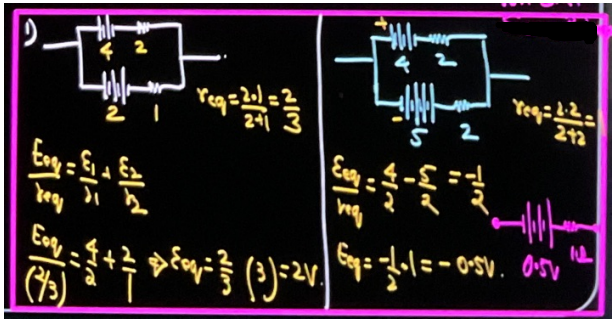
$\frac{1}{r_{eq}} = \frac{n}{r} \Rightarrow \boxed{r_{eq} = \frac{r}{n}}$

$\frac{E_{eq}}{r_{eq}} = \frac{E}{r} + \frac{E}{r} + \cdots n\text{ting}$

$\frac{E_{eq}}{r_{eq}} = \frac{nE}{r} \cdot r_{eq} = E \cdot n \left(\frac{r}{n}\right)$

$= E$

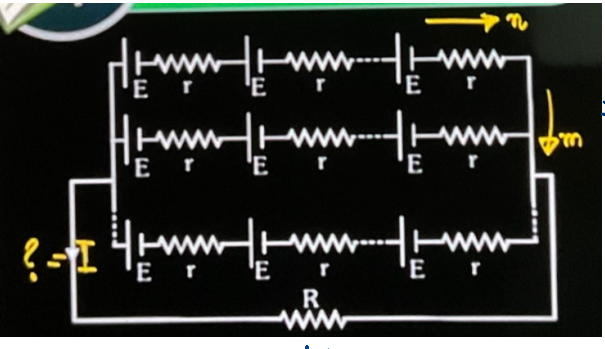


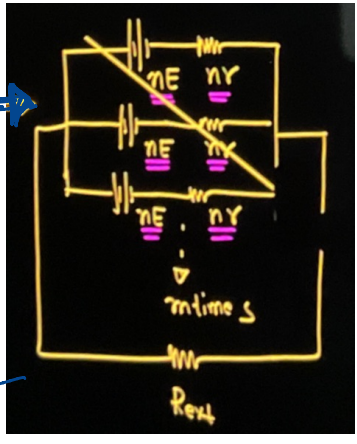


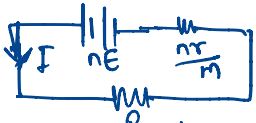
Battery grading

n identical batteries in series & m rows

- find current in Rext.







$$I = \frac{V_T}{R_T} = \frac{n\mathcal{E}}{R + \frac{nr}{m}}$$

What is the cond^n for I to be max^m –

$$I = \frac{mn\mathcal{E}}{mR + nr}$$

$I = \max^n$

deno. $\rightarrow$ min^n

$mR + nr = (\sqrt{mR})^2 + (\sqrt{nr})^2$

min^n $= (\sqrt{mR})^2 + (\sqrt{nr})^2 + 2\sqrt{mnRr} - 2\sqrt{mnRr}$

$= \underbrace{(\sqrt{mR} - \sqrt{nr})^2}_{=0} + 2\sqrt{mnRr}$

cond^n, $\sqrt{mR} = \sqrt{nr}$

$mR = nr$

$R = \frac{nr}{m}$

When R_{ext} of ckt is equal net internal resistance of batteries

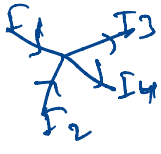
$I \rightarrow \max^m$.

Kirchoff's Laws

first law or junction law (Kirchoff's current law) (KCL)

The total current entering a junc^n or a node is equal to charge leaving the node as no charge is lost

Based on charge conservation.

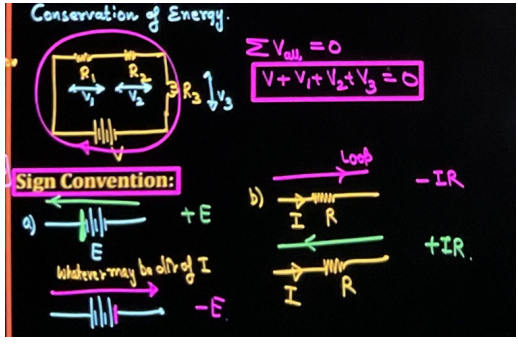


- net current at any junc^n = zero
- sign $I_{incoming}$ = -ve
  $I_{outgoing}$ = +ve

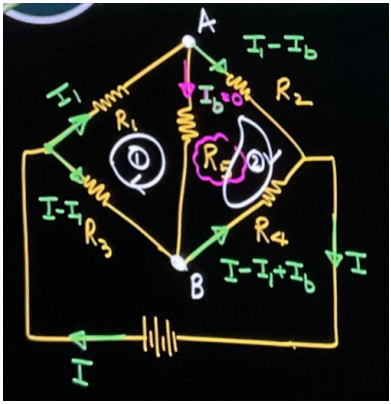
$-I_1 - I_2 + I_3 + I_4 = 0$

Second law or Junc^n law (Kirchoff's Voltage law) (KVL)

The voltage around a loop equals the sum of every voltage drop in the same loop for any closed network and equals zero



Wheatstone Bridge



Combination of 5 Resistance as shown in which if there is no current in bridging Resistance

$\rightarrow V_A = V_B \quad \rightarrow I_b = 0$

$\rightarrow \frac{R_1}{R_2} = \frac{R_3}{R_4}$

$\rightarrow$ this cond^n is c/a balanced WSB

KVL for loop 1 $+R_3 (I - I_1) - I_1 R_1 - I_b R_5 = 0$

KVL for loop 2 $+I_b R_5 - (I_1 - I_b) R_2 + (I - I_1 + I_b) R_4 = 0$

In balanced cond^n $I_b = 0$

$R_3 (I - I_1) = I_1 R_1$

$R_4 (I - I_1) = I_1 R_2$

Ratio $\Rightarrow$ $\boxed{\frac{R_1}{R_2} = \frac{R_3}{R_4}}$

Meter Bridge

— working principle = balanced cond^n of WSB.

applic^n of WSB = to find value of unknown resistance

$\Rightarrow$ It has a wire of length '1m'

when there is no deflection in Jockey ($I_g = 0$) WSB is balanced.

$$\frac{R_K}{R_M} = \frac{\rho L}{A} \Big/ \frac{\rho (100 - l)}{A}$$

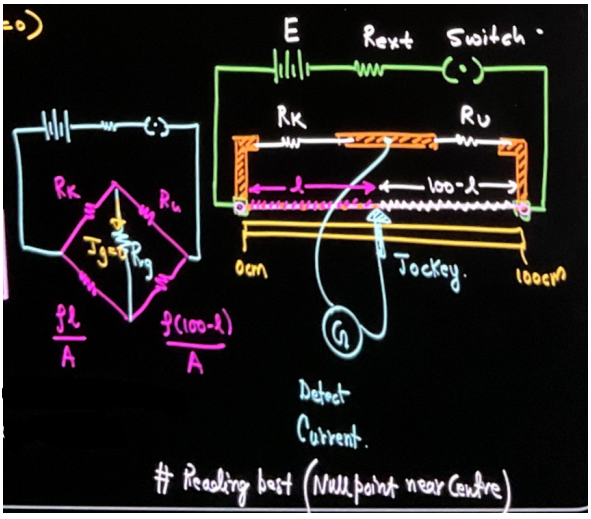
$$\frac{R_K}{R_u} = \frac{l}{100-l} \Rightarrow R_u = R_K \frac{(100-l)}{l}$$

R_K = known resistance

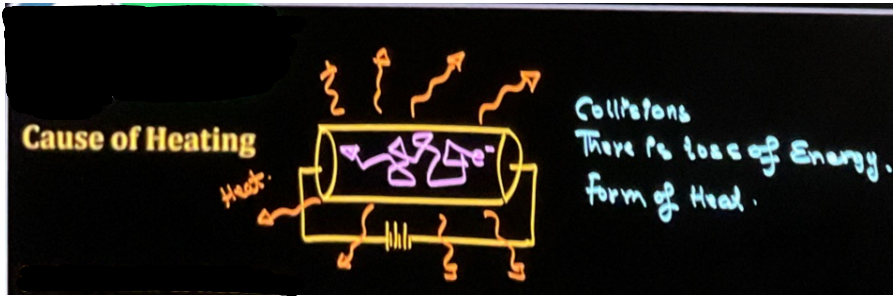
R_u = unknown resistance

end correction

wire endings at screw give additional resistance if we consider that resistance also, it is called end correction.



Heat



Joules law of heating — when a current I is made to flow through a ohmic resistance R for time t, heat H is produced such that

$$\text{heat} = I^2Rt$$
$$= \frac{V^2}{R}t$$
$$= VIt$$

Heat produced in a conductor does not depend upon the direction of current.

SI unit = J.

Power through bulbs

Power: Rate of doing work

$P = \frac{\text{Heat}}{t}$

$P = VI = \frac{V^2}{R} = I^2R$

unit = Watts.

Commercial unit = Kwh.

Series Combination I = Same, $(P = I^2R)$ better

Parallel Combination V = Same, $\left(P = \frac{V^2}{R}\right)$ better.

Bulb:

$P_0 = \frac{V_0^2}{R}$

$R_{bulb} = \frac{V_0^2}{P_0}$ → doesn't vary.

More is Power in ckt ↑ More is the Brightness

$P_{ckt} = \frac{V_{Apply}^2}{R} = \frac{V^2}{V_0^2} P_0$.

$P_{ckt} = \left(\frac{V}{V_0}\right)^2 P_0$

- $V < V_0$ Brightness ↓
- $V = V_0$ $P = P_0$.
- $V > V_0$ Brightness ↑.

Tungsten filament
- High Melting point.
- Not Easily oxidised
- R doesn't vary Rapidly Temp.

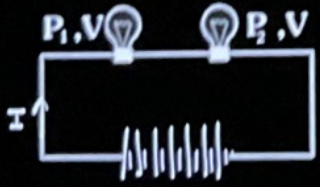
2020

P_0 = Rated Power
V_0 = Rated Voltage.

Jo us par likha hai.
Ex:- 100 watt, 220 V.
Wala 220 V milega tab (100 watt power milegi.

Brightness/Power ⇒ Replace all bulbs by $R = \frac{V_0^2}{P_0}$ then find Power in ckt, P↑ B↑, P↓ B↓.

(a) Series combinations 2024,2019

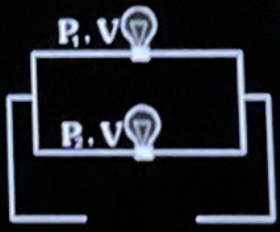


$R_S = R_1 + R_2$

$\frac{V^2}{P_S} = \frac{V^2}{P_1} + \frac{V^2}{P_2}$

$\frac{1}{P_S} = \frac{1}{P_1} + \frac{1}{P_2}$

(b) Parallel Combination 2023,2019



Parallel Comb.

$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2}$

$\frac{P_p}{V^2} = \frac{P_1}{V^2} + \frac{P_2}{V^2}$

$P_p = P_1 + P_2$

THANK YOU !!